

# Kazalo vsebine

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|                                                 |           |
|-------------------------------------------------|-----------|
| <b>1. Uvod v shemo</b>                          | <b>3</b>  |
| 1.1 Konvencije . . . . .                        | 3         |
| 1.2 Multi-tenant izolacija . . . . .            | 3         |
| 1.3 Optimistic locking . . . . .                | 3         |
| <b>2. Modul 1: Lokacije in nastavitve</b>       | <b>5</b>  |
| <b>3. Modul 2: Meniji in artikli</b>            | <b>6</b>  |
| <b>4. Modul 3: Naročila in postrežba</b>        | <b>8</b>  |
| <b>5. Modul 4: Plačila</b>                      | <b>10</b> |
| <b>6. Modul 5: Uporabniki in osebje</b>         | <b>12</b> |
| <b>7. Modul 6: FURS in računi</b>               | <b>13</b> |
| <b>8. Modul 7: Zaloge in dobavitelji</b>        | <b>15</b> |
| <b>9. Modul 8: Gosti in loyalty</b>             | <b>17</b> |
| <b>10. Modul 9: Računovodstvo</b>               | <b>18</b> |
| <b>11. Modul 10: Integracije, PWA in ostalo</b> | <b>19</b> |
| <b>12. Ključne relacije (ER)</b>                | <b>20</b> |
| 12.1 Order relacije . . . . .                   | 20        |
| 12.2 Employee relacije . . . . .                | 20        |
| 12.3 Inventory relacije . . . . .               | 20        |
| 12.4 Audit log (chain hash) . . . . .           | 20        |
| <b>13. Indeksi in performance</b>               | <b>21</b> |

|                                           |    |
|-------------------------------------------|----|
| 13.1 Ključni indeksi . . . . .            | 21 |
| 13.2 Query optimization primeri . . . . . | 21 |
| 13.3 Migracije . . . . .                  | 22 |

# 1. Uvod v shemo

RestaurantOS uporablja PostgreSQL bazo z 94 Prisma modeli (2992 vrstic sheme). Ta dokument vsebuje pregled vseh modelov razporejenih v 10 logičnih modulov z njihovimi polji, relacijami in indeksi.

|                |              |                  |                    |
|----------------|--------------|------------------|--------------------|
| 94             | 2992         | 10               | 8                  |
| Prisma modelov | vrstic sheme | logičnih modulov | multi-tenant tabel |

## 1.1 Konvencije

- **ID** - CUID (clxyz123abc...), 24 znakov, globally unique
- **Timestamps** - createdAt (auto), updatedAt (auto, za optimistic lock)
- **Soft delete** - ni privzet (hard delete z anonymize za GDPR)
- **Multi-tenant** - locationId na 8 ključnih tabelah
- **Indexes** - na vseh foreign key-ih in pogosto iskanih poljih
- **Enums** - uporabljeni za status polja (OrderStatus, PaymentStatus)

## 1.2 Multi-tenant izolacija

Naslednje tabele imajo locationId za multi-tenant izolacijo (glej ADR-004):

- **Order** - naročila so vezana na lokacijo
- **Payment** - plačila pripadajo naročilom na lokaciji
- **InventoryItem** - zaloge so ločene po lokacijah
- **Employee** - osebje je dodeljeno lokaciji
- **Table** - mize so fizično na lokaciji
- **CashRegisterShift** - izmene so po lokacijah
- **AuditLog** - log-i vsebujejo locationId kontekst
- **PushSubscription** - subscription-i so per employee/lokacija

## 1.3 Optimistic locking

Tabele z `updatedAt` poljem podpirajo optimistic locking. Pri update-u se pošlje expectedUpdatedAt - če se ne ujema s trenutnim, vrne 409 Conflict.

```
// Primer optimistic lock:
const order = await db.order.update({
  where: {
    id: orderId,
    updatedAt: expectedUpdatedAt, // preveri, da ni bil spremenjen
  },
  data: { status: 'paid' },
})
```

```
if (!order) {  
  // 409 Conflict - nekdo drug je spremenil  
  return NextResponse.json(  
    { error: 'OPTIMISTIC_LOCK_CONFLICT' },  
    { status: 409 }  
  )  
}
```

## 2. Modul 1: Lokacije in nastavitve

Osnovni modul za upravljanje lokacij (tenantov) in globalnih nastavitev restavracije.

| LocationFizična lokacija restavracije (tenant) |          |          |                              |
|------------------------------------------------|----------|----------|------------------------------|
| Field                                          | Type     | Required | Description                  |
| id                                             | String   | ✓        | CUID primary key             |
| businessId                                     | String   | ✓        | Matična številka podjetja    |
| taxId                                          | String   | ✓        | Davčna številka (za FURS)    |
| registerNumber                                 | String   | ✓        | Register number (FURS)       |
| premisesId                                     | String   | ✓        | FURS premises ID             |
| name                                           | String   | ✓        | Ime lokacije                 |
| address                                        | String   | ✓        | Naslov                       |
| fursCertPath                                   | String?  | ✗        | Pot do .p12 certifikata      |
| fursCertPassword                               | String?  | ✗        | Geslo certifikata            |
| fursEnvironment                                | String?  | ✗        | test   production            |
| isActive                                       | Boolean  | ✓        | Ali je lokacija aktivna      |
| createdAt                                      | DateTime | ✓        | Avtomatsko                   |
| updatedAt                                      | DateTime | ✓        | Avtomatsko (optimistic lock) |

| RestaurantSettingsGlobalne nastavitve restavracije |          |          |                               |
|----------------------------------------------------|----------|----------|-------------------------------|
| Field                                              | Type     | Required | Description                   |
| id                                                 | String   | ✓        | CUID                          |
| locationId                                         | String?  | ✗        | Lokacija (če je per-location) |
| name                                               | String   | ✓        | Ime restavracije              |
| currency                                           | String   | ✓        | EUR (default)                 |
| timezone                                           | String   | ✓        | Europe/Ljubljana              |
| locale                                             | String   | ✓        | sl-SI                         |
| isActive                                           | Boolean  | ✓        | Aktivne nastavitve            |
| createdAt                                          | DateTime | ✓        | Avtomatsko                    |
| updatedAt                                          | DateTime | ✓        | Avtomatsko                    |

### 3. Modul 2: Meniji in artikli

Modul za upravljanje menijev, artiklov, kategorij, modifikatorjev in cen.

| Menu       |          | Meni (npr. Kosilo, Večerja, Pijače) |                               |
|------------|----------|-------------------------------------|-------------------------------|
| Field      | Type     | Required                            | Description                   |
| id         | String   | ✓                                   | CUID                          |
| name       | String   | ✓                                   | Ime menija                    |
| locationId | String?  | ✗                                   | Lokacija (če je per-location) |
| isActive   | Boolean  | ✓                                   | Ali je meni aktiven           |
| createdAt  | DateTime | ✓                                   | Avtomatsko                    |
| updatedAt  | DateTime | ✓                                   | Avtomatsko                    |

| MenuItem        |          | Artikel v meniju (Pizza, Pivo, itd.) |                                  |
|-----------------|----------|--------------------------------------|----------------------------------|
| Field           | Type     | Required                             | Description                      |
| id              | String   | ✓                                    | CUID                             |
| name            | String   | ✓                                    | Ime artikla                      |
| nameEn          | String?  | ✗                                    | Angleško ime                     |
| price           | Decimal  | ✓                                    | Cena (EUR)                       |
| categoryId      | String   | ✓                                    | FK → Category                    |
| taxRateId       | String   | ✓                                    | FK → TaxRate (DDV stopnja)       |
| allergens       | String[] | ✗                                    | Alergeni (gluten, milk, nuts...) |
| imageUrl        | String?  | ✗                                    | URL do slike                     |
| description     | String?  | ✗                                    | Opis artikla                     |
| preparationTime | Int?     | ✗                                    | Čas priprave (minute)            |
| active          | Boolean  | ✓                                    | Ali je artikel aktiven           |
| createdAt       | DateTime | ✓                                    | Avtomatsko                       |
| updatedAt       | DateTime | ✓                                    | Avtomatsko                       |

| Category  |          | Kategorija menija (Predjedi, Glavne jedi) |                    |
|-----------|----------|-------------------------------------------|--------------------|
| Field     | Type     | Required                                  | Description        |
| id        | String   | ✓                                         | CUID               |
| name      | String   | ✓                                         | Ime kategorije     |
| menuId    | String   | ✓                                         | FK → Menu          |
| sortOrder | Int      | ✓                                         | Vrstni red prikaza |
| createdAt | DateTime | ✓                                         | Avtomatsko         |

|                      |                                            |
|----------------------|--------------------------------------------|
| <b>ModifierGroup</b> | Skupina modifikatorjev (Velikost, Dodatki) |
|----------------------|--------------------------------------------|

| Field         | Type     | Required | Description           |
|---------------|----------|----------|-----------------------|
| id            | String   | ✓        | CUID                  |
| name          | String   | ✓        | Ime skupine           |
| isRequired    | Boolean  | ✓        | Ali je obvezna izbira |
| maxSelections | Int      | ✓        | Max št. izborov       |
| createdAt     | DateTime | ✓        | Avtomatsko            |

|                 |                                               |
|-----------------|-----------------------------------------------|
| <b>Modifier</b> | Posamezni modifikator (S, M, L, Brez glutena) |
|-----------------|-----------------------------------------------|

| Field           | Type     | Required | Description                   |
|-----------------|----------|----------|-------------------------------|
| id              | String   | ✓        | CUID                          |
| name            | String   | ✓        | Ime modifikatorja             |
| modifierGroupId | String   | ✓        | FK → ModifierGroup            |
| priceModifier   | Decimal  | ✓        | Sprememba cene (+2.00, -0.50) |
| createdAt       | DateTime | ✓        | Avtomatsko                    |

|                |                             |
|----------------|-----------------------------|
| <b>TaxRate</b> | DDV stopnja (22%, 9.5%, 0%) |
|----------------|-----------------------------|

| Field     | Type     | Required | Description                  |
|-----------|----------|----------|------------------------------|
| id        | String   | ✓        | CUID                         |
| name      | String   | ✓        | Opis (npr. "Standardna 22%") |
| rate      | Decimal  | ✓        | Stopnja (22, 9.5, 0)         |
| isActive  | Boolean  | ✓        | Ali je aktivna               |
| createdAt | DateTime | ✓        | Avtomatsko                   |

## 4. Modul 3: Naročila in postrežba

Modul za upravljanje naročil, miz, artiklov v naročilu in transakcij.

| Table          |          | Miza v restavraciji |                              |
|----------------|----------|---------------------|------------------------------|
| Field          | Type     | Required            | Description                  |
| id             | String   | ✓                   | CUID                         |
| number         | Int      | ✓                   | Številka mize                |
| capacity       | Int      | ✓                   | Kapaciteta (št. oseb)        |
| section        | String?  | ✗                   | Sektor (Main Hall, Terrace)  |
| locationId     | String   | ✓                   | FK → Location (multi-tenant) |
| status         | String   | ✓                   | free   occupied   reserved   |
| currentOrderId | String?  | ✗                   | FK → Order (če je zasedena)  |
| createdAt      | DateTime | ✓                   | Avtomatsko                   |
| updatedAt      | DateTime | ✓                   | Avtomatsko                   |

| Order          |             | Naročilo (glavna entiteta) |                                  |
|----------------|-------------|----------------------------|----------------------------------|
| Field          | Type        | Required                   | Description                      |
| id             | String      | ✓                          | CUID                             |
| number         | String      | ✓                          | Številka naročila (2025-001)     |
| tableId        | String?     | ✗                          | FK → Table (če je dine-in)       |
| tableNumber    | Int?        | ✗                          | Denormalizirano za hitrost       |
| employeeId     | String      | ✓                          | FK → Employee (natakar)          |
| locationId     | String      | ✓                          | FK → Location (multi-tenant)     |
| status         | String      | ✓                          | open   paid   cancelled   storno |
| type           | String      | ✓                          | dine_in   takeaway   delivery    |
| total          | Decimal     | ✓                          | Skupni znesek                    |
| subtotal       | Decimal     | ✓                          | Pred DDV                         |
| taxTotal       | Decimal     | ✓                          | DDV skupaj                       |
| items          | OrderItem[] | ✓                          | Relacija → OrderItem             |
| payments       | Payment[]   | ✓                          | Relacija → Payment               |
| fursZoi        | String?     | ✗                          | FURS ZOI (32-znakovni MD5)       |
| fursEor        | String?     | ✗                          | FURS EOR (UUID)                  |
| fursVerifiedAt | DateTime?   | ✗                          | Datum FURS potrjevanja           |
| idempotencyKey | String?     | ✗                          | Za idempotentno ustvarjanje      |
| createdAt      | DateTime    | ✓                          | Avtomatsko                       |
| updatedAt      | DateTime    | ✓                          | Avtomatsko (optimistic lock)     |

| OrderItem  |          | Posamezni artikel v naročilu |                                      |
|------------|----------|------------------------------|--------------------------------------|
| Field      | Type     | Required                     | Description                          |
| id         | String   | ✓                            | CUID                                 |
| orderId    | String   | ✓                            | FK → Order                           |
| menuItemId | String   | ✓                            | FK → MenuItem                        |
| name       | String   | ✓                            | Denormalizirano ime                  |
| quantity   | Int      | ✓                            | Količina                             |
| price      | Decimal  | ✓                            | Cena na artikel                      |
| modifiers  | String[] | ✗                            | Izbrani modifikatorji                |
| notes      | String?  | ✗                            | Opombe (brez čebule)                 |
| status     | String   | ✓                            | pending   preparing   ready   served |
| createdAt  | DateTime | ✓                            | Avtomatsko                           |

# 5. Modul 4: Plačila

| Payment              |          | Plačilo za naročilo |                                         |
|----------------------|----------|---------------------|-----------------------------------------|
| Field                | Type     | Required            | Description                             |
| id                   | String   | ✓                   | CUID                                    |
| orderId              | String   | ✓                   | FK → Order                              |
| method               | String   | ✓                   | cash   card   gift_card                 |
| amount               | Decimal  | ✓                   | Znesek plačila                          |
| receivedAmount       | Decimal? | ✗                   | Prejeto (za gotovino)                   |
| change               | Decimal? | ✗                   | Vračilo                                 |
| tip                  | Decimal? | ✗                   | Napitnina                               |
| status               | String   | ✓                   | pending   completed   failed   refunded |
| gateway              | String?  | ✗                   | stripe   sumup   null (cash)            |
| gatewayTransactionId | String?  | ✗                   | ID transakcije pri gateway-u            |
| idempotencyKey       | String?  | ✗                   | Za idempotentno plačilo                 |
| createdAt            | DateTime | ✓                   | Avtomatsko                              |
| updatedAt            | DateTime | ✓                   | Avtomatsko                              |

| GiftCard       |           | Darilna kartica |                       |
|----------------|-----------|-----------------|-----------------------|
| Field          | Type      | Required        | Description           |
| id             | String    | ✓               | CUID                  |
| code           | String    | ✓               | Koda kartice (unique) |
| initialBalance | Decimal   | ✓               | Začetno stanje        |
| currentBalance | Decimal   | ✓               | Trenutno stanje       |
| expiresAt      | DateTime? | ✗               | Datum poteka          |
| isActive       | Boolean   | ✓               | Ali je aktivna        |
| createdAt      | DateTime  | ✓               | Avtomatsko            |

| CashRegisterShift |          | Izmena blagajne (odprtje/zapiranje) |                              |
|-------------------|----------|-------------------------------------|------------------------------|
| Field             | Type     | Required                            | Description                  |
| id                | String   | ✓                                   | CUID                         |
| locationId        | String   | ✓                                   | FK → Location (multi-tenant) |
| employeeId        | String   | ✓                                   | FK → Employee (odprl)        |
| openingAmount     | Decimal  | ✓                                   | Začetno stanje gotovine      |
| closingAmount     | Decimal? | ✗                                   | Končno stanje                |
| countedCash       | Decimal? | ✗                                   | Prešteto stanje              |

| Field       | Type      | Required | Description   |
|-------------|-----------|----------|---------------|
| discrepancy | Decimal?  | ✖        | Razlika       |
| status      | String    | ✓        | open   closed |
| openedAt    | DateTime  | ✓        | Čas odprtja   |
| closedAt    | DateTime? | ✖        | Čas zaprtja   |

# 6. Modul 5: Uporabniki in osebe

| Employee     |           | Zaposleni (natakar, kuhar, manager) |                                               |
|--------------|-----------|-------------------------------------|-----------------------------------------------|
| Field        | Type      | Required                            | Description                                   |
| id           | String    | ✓                                   | CUID                                          |
| name         | String    | ✓                                   | Ime in priimek                                |
| pin          | String    | ✓                                   | Hashed PIN (bcrypt + HMAC)                    |
| role         | String    | ✓                                   | waiter   cook   manager   admin   super_admin |
| locationId   | String    | ✓                                   | FK → Location (multi-tenant)                  |
| email        | String?   | ✗                                   | Email (unique)                                |
| phone        | String?   | ✗                                   | Telefon (E.164 format)                        |
| active       | Boolean   | ✓                                   | Ali je aktivni                                |
| anonymizedAt | DateTime? | ✗                                   | GDPR izbris (anonymize)                       |
| createdAt    | DateTime  | ✓                                   | Avtomatsko                                    |
| updatedAt    | DateTime  | ✓                                   | Avtomatsko                                    |

| Session    |          | Aktivna seja uporabnika |                     |
|------------|----------|-------------------------|---------------------|
| Field      | Type     | Required                | Description         |
| id         | String   | ✓                       | CUID                |
| employeeId | String   | ✓                       | FK → Employee       |
| token      | String   | ✓                       | JWT token (unique)  |
| expiresAt  | DateTime | ✓                       | Datum poteka (8h)   |
| ipAddress  | String?  | ✗                       | IP naslov prijave   |
| userAgent  | String?  | ✗                       | Browser/device info |
| createdAt  | DateTime | ✓                       | Avtomatsko          |

| PushSubscription |          | Web Push subscription |                      |
|------------------|----------|-----------------------|----------------------|
| Field            | Type     | Required              | Description          |
| id               | String   | ✓                     | CUID                 |
| employeeId       | String   | ✓                     | FK → Employee        |
| endpoint         | String   | ✓                     | Push endpoint URL    |
| keysP256dh       | String   | ✓                     | P256dh key           |
| keysAuth         | String   | ✓                     | Auth key             |
| expirationTime   | Int?     | ✗                     | Expiration timestamp |
| createdAt        | DateTime | ✓                     | Avtomatsko           |

# 7. Modul 6: FURS in računi

| ReceiptRačun (FURS potrjen) |           |          |                                 |
|-----------------------------|-----------|----------|---------------------------------|
| Field                       | Type      | Required | Description                     |
| id                          | String    | ✓        | CUID                            |
| orderId                     | String    | ✓        | FK → Order                      |
| receiptNumber               | String    | ✓        | Številka računa (unique)        |
| zoi                         | String    | ✓        | FURS ZOI (32-znakovni MD5)      |
| eor                         | String?   | ✗        | FURS EOR (UUID, po potrjevanju) |
| fursVerifiedAt              | DateTime? | ✗        | Datum FURS potrjevanja          |
| qrCode                      | String?   | ✗        | Base64 QR koda                  |
| pdfUrl                      | String?   | ✗        | URL do PDF računa               |
| stornoReceiptId             | String?   | ✗        | FK → Receipt (če je storno)     |
| createdAt                   | DateTime  | ✓        | Avtomatsko                      |

| ZReportZ-Report (dnevno zapiranje) |          |          |                       |
|------------------------------------|----------|----------|-----------------------|
| Field                              | Type     | Required | Description           |
| id                                 | String   | ✓        | CUID                  |
| locationId                         | String   | ✓        | FK → Location         |
| zReportNumber                      | String   | ✓        | Številka (Z-2025-001) |
| openingAmount                      | Decimal  | ✓        | Začetno stanje        |
| closingAmount                      | Decimal  | ✓        | Končno stanje         |
| cashSales                          | Decimal  | ✓        | Gotovinska prodaja    |
| cardSales                          | Decimal  | ✓        | Kartična prodaja      |
| totalSales                         | Decimal  | ✓        | Skupna prodaja        |
| discrepancy                        | Decimal  | ✓        | Razlika               |
| closedAt                           | DateTime | ✓        | Čas zaprtja           |

| AuditLogNepopravljiv audit log (chain hash) |          |          |                                           |
|---------------------------------------------|----------|----------|-------------------------------------------|
| Field                                       | Type     | Required | Description                               |
| id                                          | String   | ✓        | CUID                                      |
| timestamp                                   | DateTime | ✓        | Avtomatsko                                |
| employeeId                                  | String   | ✓        | FK → Employee                             |
| action                                      | String   | ✓        | furs.verify   payment.create   auth.login |
| entityId                                    | String   | ✓        | ID entitete, na katero se nanaša          |
| entityType                                  | String   | ✓        | order   payment   employee                |

| Field        | Type    | Required | Description                     |
|--------------|---------|----------|---------------------------------|
| ipAddress    | String? | ✘        | IP naslov                       |
| metadata     | Json?   | ✘        | Dodatni podatki                 |
| previousHash | String  | ✓        | Hash prejšnje vrstice           |
| currentHash  | String  | ✓        | SHA-256(previousHash + content) |

# 8. Modul 7: Zaloge in dobavitelji

| InventoryItem |          | Artikel v zalogi |                              |
|---------------|----------|------------------|------------------------------|
| Field         | Type     | Required         | Description                  |
| id            | String   | ✓                | CUID                         |
| name          | String   | ✓                | Ime artikla                  |
| quantity      | Decimal  | ✓                | Trenutna količina            |
| unit          | String   | ✓                | kg   l   kos   g             |
| minQuantity   | Decimal  | ✓                | Minimalna količina (alert)   |
| maxQuantity   | Decimal? | ✗                | Maksimalna količina          |
| cost          | Decimal  | ✓                | Cena na enoto                |
| locationId    | String   | ✓                | FK → Location (multi-tenant) |
| supplierId    | String?  | ✗                | FK → Supplier                |
| createdAt     | DateTime | ✓                | Avtomatsko                   |
| updatedAt     | DateTime | ✓                | Avtomatsko                   |

| Supplier  |          | Dobavitelj |                 |
|-----------|----------|------------|-----------------|
| Field     | Type     | Required   | Description     |
| id        | String   | ✓          | CUID            |
| name      | String   | ✓          | Ime dobavitelja |
| taxId     | String?  | ✗          | Davčna številka |
| email     | String?  | ✗          | Email           |
| phone     | String?  | ✗          | Telefon         |
| address   | String?  | ✗          | Naslov          |
| isActive  | Boolean  | ✓          | Ali je aktiven  |
| createdAt | DateTime | ✓          | Avtomatsko      |

| PurchaseOrder |          | Nabavni order |                                     |
|---------------|----------|---------------|-------------------------------------|
| Field         | Type     | Required      | Description                         |
| id            | String   | ✓             | CUID                                |
| supplierId    | String   | ✓             | FK → Supplier                       |
| locationId    | String   | ✓             | FK → Location                       |
| orderNumber   | String   | ✓             | Številka naročila                   |
| status        | String   | ✓             | draft   sent   received   cancelled |
| total         | Decimal  | ✓             | Skupni znesek                       |
| orderedAt     | DateTime | ✓             | Datum naročila                      |

| Field      | Type      | Required | Description   |
|------------|-----------|----------|---------------|
| receivedAt | DateTime? | ✖        | Datum prejema |

# 9. Modul 8: Gosti in loyalty

| Guest       |          | Gost (stranka) |                   |
|-------------|----------|----------------|-------------------|
| Field       | Type     | Required       | Description       |
| id          | String   | ✓              | CUID              |
| name        | String?  | ✗              | Ime (če je znano) |
| email       | String?  | ✗              | Email (unique)    |
| phone       | String?  | ✗              | Telefon           |
| locationId  | String?  | ✗              | Primarna lokacija |
| totalVisits | Int      | ✓              | Št. obiskov       |
| totalSpent  | Decimal  | ✓              | Skupna poraba     |
| createdAt   | DateTime | ✓              | Avtomatsko        |

| LoyaltyAccount |          | Loyalty račun gosta |                                   |
|----------------|----------|---------------------|-----------------------------------|
| Field          | Type     | Required            | Description                       |
| id             | String   | ✓                   | CUID                              |
| guestId        | String   | ✓                   | FK → Guest                        |
| points         | Int      | ✓                   | Trenutno stanje točk              |
| tier           | String   | ✓                   | bronze   silver   gold   platinum |
| joinedAt       | DateTime | ✓                   | Datum pridružitve                 |

| Reservation |          | Rezervacija mize |                                                    |
|-------------|----------|------------------|----------------------------------------------------|
| Field       | Type     | Required         | Description                                        |
| id          | String   | ✓                | CUID                                               |
| guestId     | String?  | ✗                | FK → Guest (če je registriran)                     |
| guestName   | String   | ✓                | Ime gosta                                          |
| guestPhone  | String   | ✓                | Telefon                                            |
| tableId     | String?  | ✗                | FK → Table (če je dodeljena)                       |
| locationId  | String   | ✓                | FK → Location                                      |
| partySize   | Int      | ✓                | Št. oseb                                           |
| reservedAt  | DateTime | ✓                | Datum in ura rezervacije                           |
| status      | String   | ✓                | pending   confirmed   seated   cancelled   no_show |
| notes       | String?  | ✗                | Opombe                                             |
| createdAt   | DateTime | ✓                | Avtomatsko                                         |

# 10. Modul 9: Računovodstvo

| JournalEntry |               | Računovodski vnos |                             |
|--------------|---------------|-------------------|-----------------------------|
| Field        | Type          | Required          | Description                 |
| id           | String        | ✓                 | CUID                        |
| entryNumber  | String        | ✓                 | Številka vnosa              |
| date         | DateTime      | ✓                 | Datum vnosa                 |
| description  | String        | ✓                 | Opis                        |
| reference    | String?       | ✗                 | Referenca (naročilo, račun) |
| lines        | JournalLine[] | ✓                 | Relacija → JournalLine      |
| status       | String        | ✓                 | draft   posted   reversed   |
| createdAt    | DateTime      | ✓                 | Avtomatsko                  |

| JournalLine    |         | Posamezna vrstica journal entry-ja |                     |
|----------------|---------|------------------------------------|---------------------|
| Field          | Type    | Required                           | Description         |
| id             | String  | ✓                                  | CUID                |
| journalEntryId | String  | ✓                                  | FK → JournalEntry   |
| accountId      | String  | ✓                                  | FK → ChartOfAccount |
| debit          | Decimal | ✓                                  | Breme (0 če kredit) |
| credit         | Decimal | ✓                                  | Kredit (0 če breme) |
| description    | String? | ✗                                  | Opis vrstice        |

| ChartOfAccount |         | Kontni plan |                                                |
|----------------|---------|-------------|------------------------------------------------|
| Field          | Type    | Required    | Description                                    |
| id             | String  | ✓           | CUID                                           |
| code           | String  | ✓           | Konto (npr. 1200 - Blagajna)                   |
| name           | String  | ✓           | Ime konta                                      |
| type           | String  | ✓           | asset   liability   equity   revenue   expense |
| isActive       | Boolean | ✓           | Ali je aktiven                                 |

## 11. Modul 10: Integracije, PWA in ostalo

| Webhook <span>Webhook endpoint (zunanji sistemi)</span> |          |          |                                               |
|---------------------------------------------------------|----------|----------|-----------------------------------------------|
| Field                                                   | Type     | Required | Description                                   |
| id                                                      | String   | ✓        | CUID                                          |
| url                                                     | String   | ✓        | URL webhook-a                                 |
| secret                                                  | String   | ✓        | HMAC secret za signature                      |
| events                                                  | String[] | ✓        | Event tipi (order.created, payment.completed) |
| isActive                                                | Boolean  | ✓        | Ali je aktiven                                |
| createdAt                                               | DateTime | ✓        | Avtomatsko                                    |

| OutboxEvent <span>Outbox pattern za async procesiranje</span> |           |          |                                       |
|---------------------------------------------------------------|-----------|----------|---------------------------------------|
| Field                                                         | Type      | Required | Description                           |
| id                                                            | String    | ✓        | CUID                                  |
| aggregateType                                                 | String    | ✓        | order   payment   invoice             |
| aggregateId                                                   | String    | ✓        | ID entitete                           |
| type                                                          | String    | ✓        | Event tip (created, updated, deleted) |
| payload                                                       | Json      | ✓        | Event podatki                         |
| processedAt                                                   | DateTime? | ✗        | Datum obdelave                        |
| createdAt                                                     | DateTime  | ✓        | Avtomatsko                            |

| Counter <span>Števec (za generacijo števil)</span> |         |          |                                           |
|----------------------------------------------------|---------|----------|-------------------------------------------|
| Field                                              | Type    | Required | Description                               |
| id                                                 | String  | ✓        | CUID                                      |
| key                                                | String  | ✓        | Tip števca (order_number, receipt_number) |
| value                                              | Int     | ✓        | Trenutna vrednost                         |
| locationId                                         | String? | ✗        | Lokacija (če je per-location)             |
| year                                               | Int     | ✓        | Leto (za reset)                           |

## 12. Ključne relacije (ER)

---

Pregled ključnih relacij med modeli. Order je centralna entiteta, na katero se nanaša večina drugih.

### 12.1 Order relacije

Order (central)

- └ Table (1:N - miza ima lahko več naročil)
- └ Employee (N:1 - natakar ustvari naročilo)
- └ Location (N:1 - multi-tenant)
- └ OrderItem (1:N - naročilo ima več artiklov)
  - └ MenuItem (N:1 - artikel iz menija)
- └ Payment (1:N - lahko več plačil za split bill)
  - └ GiftCard (N:1 - če plačilo z darilno kartico)
- └ Receipt (1:1 - FURS račun)
- └ DeliveryInfo (1:1 - če je dostava)
- └ Discount (1:N - popusti na naročilu)

### 12.2 Employee relacije

Employee (osebje)

- └ Location (N:1 - multi-tenant)
- └ Session (1:N - aktivne seje)
- └ PushSubscription (1:N - push subscriptions)
- └ Order (1:N - naročila, ki jih je ustvaril)
- └ CashRegisterShift (1:N - izmene, ki jih je odprl)
- └ AuditLog (1:N - akcije, ki jih je izvedel)
- └ EmployeeJob (N:M - posluje preko join tabele)
- └ Job (M:N - vloge/posli')

### 12.3 Inventory relacije

InventoryItem (zaloga)

- └ Location (N:1 - multi-tenant)
- └ Supplier (N:1 - dobavitelj)
- └ StockTransaction (1:N - transakcije)
- └ PurchaseOrderItem (1:N - v nabavnih naročilih)
- └ RecipeItem (1:N - v recepturah)
- └ ReorderRule (1:1 - pravilo za avtomatsko naročanje)

### 12.4 Audit log (chain hash)

AuditLog (nepopravljiv)

- └ Employee (N:1 - kdo je izvedel akcijo)
- └ previousHash → AuditLog.currentHash (chain)
- └ currentHash = SHA-256(previousHash + content)

Verifikacija:

- Rechain od genesis do zadnje vrstice
- Če se kak hash ne ujema → COMPROMISED
- Endpoint: GET /api/audit-log/verify

## 13. Indeksi in performance

Strategija indeksov za optimalno delovanje. Vsi foreign key-i so indeksirani, prav tako pogosto iskana polja.

### 13.1 Ključni indeksi

| Tabela           | Indeks                        | Polja                   | Namen                         |
|------------------|-------------------------------|-------------------------|-------------------------------|
| Order            | idx_orders_location_created   | (locationId, createdAt) | Pregled po lokaciji in datumu |
| Order            | idx_orders_status             | (status)                | Filter po statusu (open/paid) |
| Order            | idx_orders_employee           | (employeeId)            | Pregled po natakarju          |
| Payment          | idx_payments_gateway_txid     | (gatewayTransactionId)  | Stripe webhook lookup         |
| Payment          | idx_payments_order            | (orderId)               | Plačila za naročilo           |
| AuditLog         | idx_audit_timestamp           | (timestamp)             | Časovni pregled               |
| AuditLog         | idx_audit_employee            | (employeeId)            | Pregled po uporabniku         |
| InventoryItem    | idx_inventory_location        | (locationId)            | Multi-tenant filter           |
| Employee         | idx_employees_location_active | (locationId, active)    | Aktivni uporabniki            |
| PushSubscription | idx_push_employee             | (employeeId)            | Pošiljanje push               |
| MenuItem         | idx_menu_category             | (categoryId)            | Artikli po kategoriji         |
| Receipt          | idx_receipt_zoi               | (zoi)                   | FURS ZOI lookup               |

Tabela 13.1: Ključni indeksi

### 13.2 Query optimization primeri

```
# ❌ PRAVILNO - z index-om:  
const orders = await db.order.findMany({  
  where: {  
    locationId: locationId, // index: idx_orders_location_created  
    createdAt: { gte: dateFrom },  
  },  
});
```

```

    orderBy: { createdAt: 'desc' },
    take: 20,
  })

# ✅ PRAVILNO - z include (1 query, ne N+1):
const order = await db.order.findUnique({
  where: { id: orderId },
  include: {
    items: true,      // 1 query
    payments: true,   // 1 query
    table: true,      // 1 query
    employee: {       // 1 query
      select: { name: true } // samo potrebna polja
    },
  },
})

# ❌ NAROBE - N+1 problem:
const orders = await db.order.findMany()
for (const order of orders) {
  const items = await db.orderItem.findMany({ // N query-jev!
    where: { orderId: order.id }
  })
}

```

## 13.3 Migracije

```

# Ustvari migracijo iz spremembe sheme:
bun run db:migrate -- --name add_new_field

# Apply na bazo:
bun run db:push # dev (brez migracije)
bun run db:migrate deploy # produkcija

# Reset baze (pazi!):
bun run db:reset

# Prisma Studio (GUI za DB):
bunx prisma studio
# Odpri http://localhost:5555

```

### DATABASE BEST PRACTICES

1. Vedno uporabi `include` za relacije (prepreči N+1). 2. Dodaj index na pogosto filtrirana polja. 3. Uporabi `select` za samo potrebna polja (manj podatkov). 4. Optimistic locking z `updatedAt`. 5. Multi-tenant: vedno filtriraj z locationId.